

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

### ***Device Specifications:***

Device Type: Video Projector  
 Manufacturer: Epson  
 Firmware Version: V200  
 Model(s): EB-475W, EB-475Wi, EB-485W, EB-485Wi, EB-425W, EB-435W, BrightLink 475Wi, BrightLink 480i, BrightLink 485Wi, BrightLink 575Wi, EB-420, EB-430, EB-421i, EB-431i, EB-426Wi, EB-436Wi, BrightLink 585Wi, BrightLink 595Wi, EB-575Wi, EB-585Wi, EB-595Wi, EB-CU610Wi, EB-CU610Xi, EB-CU610X, EB-CU610W, EB-CU600Wi, EB-CU600W, EB-CU600Xi, EB-CU600X, PowerLite 425W, PowerLite 435W, EB-CS500WN, EB-CS520WN, PowerLite 420, PowerLite 430, EB-CS500XN, EB-CS510XN, PowerLite 470, PowerLite 480, PowerLite CU600X, PowerLite CU610X, BrightLink 425Wi, BrightLink 430i, BrightLink 435Wi, EB-480i, BrightLink 475W, EB-485WT, BrightLink 485Wi+, BrightLink 475Wi+, PowerLite 485W, EB-470, EB-480, EB-475We, EB-475Wie, EB-485We, EB-480e, EB-480T, EB-470i, EB-485Wie, EB-425We, EB-435We, BrightLink 536Wi

### ***Tested on the Following Software and Firmware Versions:***

| IP Link Pro Control Processor Firmware | Touch Link Panel Firmware | GS Version |
|----------------------------------------|---------------------------|------------|
| 2.04.0000-b007                         | 2.01.0001-b001            | 1.2.2      |

### ***Version History***

| Module Version | Date       | Notes                                                                                    |
|----------------|------------|------------------------------------------------------------------------------------------|
| 1_1_1_0        | 2/7/2017   | Fixed Input command for BrightLink 575Wi, BrightLink 585Wi, and BrightLink 595Wi models. |
| 1_1_0_0        | 12/22/2016 | Fixed Aspect Ratio.                                                                      |
| 1_0_3_1        | 10/18/2016 | Added initialization logic                                                               |

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|         |           |                 |
|---------|-----------|-----------------|
| 1_0_3_0 | 8/31/2016 | Initial Version |
|---------|-----------|-----------------|

**Module Notes**

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example: InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example: InterfaceName.connectionCounter = 5
- Standby Mode (in Extended Tab of OSD Menu) must be set to Communication ON for the module to function correctly.
- For Serial Over Ethernet support, SerialOverEthernetClass must be used. Example: InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', PortNum, Model='ModelName')

**Control Commands, States, and Qualifiers:**

|                                      |                                                                                                                                                                                     |                                                                                                                                 |                                                                                                                     |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>AVMute</b>                        | 'On'                                                                                                                                                                                | 'Off'                                                                                                                           |                                                                                                                     |
| <b>AspectRatio</b>                   | 'Normal'<br>'Zoom' <sup>1</sup><br>'4:3' <sup>2</sup>                                                                                                                               | '16:9'<br>'Native' <sup>1</sup>                                                                                                 | 'Full' <sup>1</sup><br>'Auto'                                                                                       |
| <b>AutoImage</b>                     | None                                                                                                                                                                                |                                                                                                                                 |                                                                                                                     |
| <b>CeilingMode</b>                   | 'On'                                                                                                                                                                                | 'Off'                                                                                                                           |                                                                                                                     |
| <b>ClosedCaption</b>                 | 'CC2'                                                                                                                                                                               | 'CC1'                                                                                                                           | 'Off'                                                                                                               |
| <b>Freeze</b>                        | 'On'                                                                                                                                                                                | 'Off'                                                                                                                           |                                                                                                                     |
| <b>Input</b>                         | 'Computer 1 RGB'<br>'Computer 2 YPbPr' <sup>6</sup><br>'USB Display'<br>'HDMI 1' <sup>4</sup><br>'Computer 1 Auto' <sup>3</sup><br>'RGB-Video' <sup>3</sup><br>'Video' <sup>3</sup> | 'Computer 1 YPbPr'<br>'Composite'<br>'USB'<br>'HDMI 2' <sup>4</sup><br>'Computer 2 Auto' <sup>3,6</sup><br>'YCbCr' <sup>3</sup> | 'Computer 2 RGB' <sup>6</sup><br>'S-Video'<br>'LAN'<br>'HDMI'<br>'Digital-RGB' <sup>3</sup><br>'YPbPr' <sup>3</sup> |
| <b>LampMode</b>                      | 'Normal'                                                                                                                                                                            | 'Eco'                                                                                                                           |                                                                                                                     |
| <b>MenuNavigation</b>                | 'Menu'<br>'Down'                                                                                                                                                                    | 'Enter'<br>'Left'                                                                                                               | 'Up'<br>'Right'                                                                                                     |
| <b>Power</b>                         | 'On'                                                                                                                                                                                | 'Standby'                                                                                                                       |                                                                                                                     |
| <b>RearMode</b>                      | 'On'                                                                                                                                                                                | 'Off'                                                                                                                           |                                                                                                                     |
| <b>TouchUnitControl</b> <sup>5</sup> | 'On'                                                                                                                                                                                | 'Off'                                                                                                                           |                                                                                                                     |
| <b>Volume</b>                        | 0 to 20 in steps of 1                                                                                                                                                               |                                                                                                                                 |                                                                                                                     |

<sup>1</sup> WXGA Models support these additional Aspect Ratio states.

<sup>2</sup> XGA Models support these additional Aspect Ratio states.

<sup>3</sup> Models EB(BrightLink)-485Wi, EB(BrightLink)-475W, EB(BrightLink)-475Wi, EB(BrightLink)-480i, EB-485WT, EB-485Wie, BrightLink 485Wi+, BrightLink 475Wi+, EB(PL)-485W, EB(PL)-470, EB(PL)-480, EB-475We, EB-475Wie, EB-485We, EB-CU600W, EB-CU610Wi, EB-CU610W, EB(PL)-CU600X, EB-CU610X, EB-CU610Xi, EB-480e, EB-480T, EB-470i, EB(PL)-420, EB-425W, EB-430, EB-435W, EB-425We, 435We, PL-425W, BrightLink 425Wi, EB-430, EB(PL)-435W, EB-CS500XN, EB-CS500WN, EB-CS510XN, and EB-CS520WN have these additional inputs.

<sup>4</sup> Models BrightLink 575Wi, BrightLink 585Wi, BrightLink 595Wi, EB-575Wi, EB-585Wi, EB-595Wi have 2 HDMI inputs.

<sup>5</sup> This command is only supported by the BrightLink 595Wi.

<sup>6</sup> Models BrightLink 575Wi and BrightLink 585Wi do not support Computer 2 Inputs

**Status Available:**

|                                      |                                                                                                                                                                                                        |                                                                                                                                                                                          |                                                                                                                                                                                                                                                                       |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AVMute</b>                        | 'On'                                                                                                                                                                                                   | 'Off'                                                                                                                                                                                    |                                                                                                                                                                                                                                                                       |
| <b>AspectRatio</b>                   | 'Normal'<br>'Zoom' <sup>1</sup><br>'4:3' <sup>2</sup>                                                                                                                                                  | '16:9'<br>'Native' <sup>1</sup>                                                                                                                                                          | 'Full' <sup>1</sup><br>'Auto'                                                                                                                                                                                                                                         |
| <b>CeilingMode</b>                   | 'On'                                                                                                                                                                                                   | 'Off'                                                                                                                                                                                    |                                                                                                                                                                                                                                                                       |
| <b>ClosedCaption</b>                 | 'CC2'                                                                                                                                                                                                  | 'CC1'                                                                                                                                                                                    | 'Off'                                                                                                                                                                                                                                                                 |
| <b>ConnectionStatus</b>              | 'Connected'                                                                                                                                                                                            | 'Disconnected'                                                                                                                                                                           |                                                                                                                                                                                                                                                                       |
| <b>DeviceStatus</b> <sup>6</sup>     | 'Normal'<br><br>'High Internal Temperature Error'<br>'Cinema Filter Error'<br>'Subsystem Error'<br><br>'Power Supply Unit Error (Ballast)'<br>'Cooling System Error (Pump)'<br>'Exhaust Shutter Error' | 'Fan Error'<br><br>'Lamp Error'<br>'Electric Dual-Layered Capacitor Is Disconnected'<br>'Low Air Flow Error'<br><br>'Shutter Error'<br>'Static Iris Error'<br>'Obstacle Detection Error' | 'Lamp Failure At Power On'<br><br>'Open Lamp Cover Door Error'<br>'Auto Iris Error'<br><br>'Air Filter Air Flow Sensor Error'<br><br>'Cooling System Error (Peltiert Element)'<br>'Power Supply Unit Error (Disagreement Of Ballast)'<br>'IF Board Discernment Error' |
| <b>Freeze</b>                        | 'On'                                                                                                                                                                                                   | 'Off'                                                                                                                                                                                    |                                                                                                                                                                                                                                                                       |
| <b>Input</b>                         | 'Computer 1 RGB'<br>'Computer 2 YPbPr' <sup>7</sup><br>'USB Display'<br>'HDMI 1' <sup>4</sup><br>'Computer 1 Auto' <sup>3</sup><br>'RGB-Video' <sup>3</sup><br>'Video' <sup>3</sup>                    | 'Computer 1 YPbPr'<br>'Composite'<br>'USB'<br>'HDMI 2' <sup>4</sup><br>'Computer 2 Auto' <sup>3, 7</sup><br>'YCbCr' <sup>3</sup>                                                         | 'Computer 2 RGB' <sup>7</sup><br>'S-Video'<br>'LAN'<br>'HDMI'<br>'Digital-RGB' <sup>3</sup><br>'YPbPr' <sup>3</sup>                                                                                                                                                   |
| <b>LampMode</b>                      | 'Normal'                                                                                                                                                                                               | 'Eco'                                                                                                                                                                                    |                                                                                                                                                                                                                                                                       |
| <b>LampUsage</b>                     | Hours                                                                                                                                                                                                  |                                                                                                                                                                                          |                                                                                                                                                                                                                                                                       |
| <b>Power</b>                         | 'On'<br>'Cooling Down'<br>'A/V Standby'                                                                                                                                                                | 'Standby'<br>'Network Standby'                                                                                                                                                           | 'Warming Up'<br>'Abnormal Standby'                                                                                                                                                                                                                                    |
| <b>RearMode</b>                      | 'On'                                                                                                                                                                                                   | 'Off'                                                                                                                                                                                    |                                                                                                                                                                                                                                                                       |
| <b>TouchUnitControl</b> <sup>5</sup> | 'On'                                                                                                                                                                                                   | 'Off'                                                                                                                                                                                    |                                                                                                                                                                                                                                                                       |

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|               |                       |
|---------------|-----------------------|
| <b>Volume</b> | 0 to 20 in steps of 1 |
|---------------|-----------------------|

<sup>1</sup> WXGA Models support these additional Aspect Ratio states.

<sup>2</sup> XGA Models support these additional Aspect Ratio states.

<sup>3</sup> Models EB(BrightLink)-485Wi, EB(BrightLink)-475W, EB(BrightLink)-475Wi, EB(BrightLink)-480i, EB-485WT, EB-485Wie, BrightLink 485Wi+, BrightLink 475Wi+, EB(PL)-485W, EB(PL)-470, EB(PL)-480, EB-475We, EB-475Wie, EB-485We, EB-CU600W, EB-CU610Wi, EB-CU610W, EB(PL)-CU600X, EB-CU610X, EB-CU610Xi, EB-480e, EB-480T, EB-470i, EB(PL)-420, EB-425W, EB-430, EB-435W, EB-425We, 435We, PL-425W, BrightLink 425Wi, EB-430, EB(PL)-435W, EB-CS500XN, EB-CS500WN, EB-CS510XN, and EB-CS520WN have these additional inputs.

<sup>4</sup> Models BrightLink 575Wi, BrightLink 585Wi, BrightLink 595Wi, EB-575Wi, EB-585Wi, EB-595Wi have 2 HDMI inputs.

<sup>5</sup> This command is only supported by the BrightLink 595Wi.

<sup>6</sup> Only supported via serial connection.

<sup>7</sup> Models BrightLink 575Wi and BrightLink 585Wi do not support Computer 2 Inputs.

## ***Cable and Adapter Requirements:***

Captive Screw to Female DB9 RS-232 Serial Cable

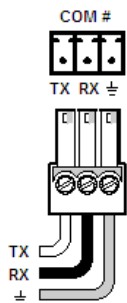
## ***Notes for the Device:***

## ***Serial communication:***

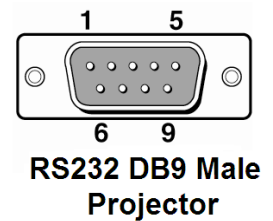
**Port Type:** RS-232  
**Baud Rate:** 9600  
**Data Bits:** 8

**Parity:** None  
**Stop Bits:** One  
**Flow Control:** None

## ***Pin Assignments Diagram***



| Signal | Main Cable | Pin | Signal |
|--------|------------|-----|--------|
| TxD    | →          | 2   | RxD    |
| RxD    | ←          | 3   | TxD    |
| GND    | →          | 5   | GND    |



## ***General Notes:***

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***Network communication:***

When configuring the Ethernet interface, be sure the device settings match that of the GS interface settings.

**Port Type:** Ethernet

**Default Port:** 3629

**Logon Credentials** No

**Supported:**

**Multi-Connection** Yes

**Capabilities:**

**Port Changeability:** No

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***Ethernet Module Configuration Description:***

Please refer to user manual for settings and changes to the network communication

***Notes for the Device:***

## Appendix A. Set Commands

|                        |                  |
|------------------------|------------------|
| A/V Mute Off           | MUTE OFF\x0D     |
| A/V Mute On            | MUTE ON\x0D      |
| Aspect Ratio 16:9      | ASPECT 20\x0D    |
| Aspect Ratio 4:3       | ASPECT 10\x0D    |
| Aspect Ratio Auto      | ASPECT 30\x0D    |
| Aspect Ratio Full      | ASPECT 40\x0D    |
| Aspect Ratio Native    | ASPECT 60\x0D    |
| Aspect Ratio Normal    | ASPECT 00\x0D    |
| Aspect Ratio Zoom      | ASPECT 50\x0D    |
| Auto Image None        | KEY 4A\x0D       |
| Ceiling Mode Off       | VREVERSE OFF\x0D |
| Ceiling Mode On        | VREVERSE ON\x0D  |
| Closed Caption CC1     | CCAP 11\x0D      |
| Closed Caption CC2     | CCAP 12\x0D      |
| Closed Caption Off     | CCAP 00\x0D      |
| Freeze Off             | FREEZE OFF\x0D   |
| Freeze On              | FREEZE ON\x0D    |
| Input Composite        | SOURCE 41\x0D    |
| Input Computer 1 RGB   | SOURCE 11\x0D    |
| Input Computer 1 YPbPr | SOURCE 14\x0D    |
| Input Computer 2 RGB   | SOURCE 21\x0D    |
| Input Computer 2 YPbPr | SOURCE 24\x0D    |
| Input LAN              | SOURCE 53\x0D    |
| Input S-Video          | SOURCE 42\x0D    |
| Input USB              | SOURCE 52\x0D    |
| Input USB Display      | SOURCE 51\x0D    |
| Input HDMI 1           | SOURCE 30\x0D    |
| Input HDMI 2           | SOURCE A0\x0D    |
| Input RGB-Video        | SOURCE 33\x0D    |
| Input Digital-RGB      | SOURCE 31\x0D    |
| Input Video            | SOURCE 40\x0D    |
| Input YCbCr            | SOURCE 34\x0D    |
| Input YPbPr            | SOURCE 35\x0D    |
| Input HDMI             | SOURCE 30\x0D    |
| Lamp Mode Eco          | LUMINANCE 01\x0D |
| Lamp Mode Normal       | LUMINANCE 00\x0D |
| Menu Navigation Down   | KEY 36\x0D       |
| Menu Navigation Enter  | KEY 16\x0D       |
| Menu Navigation Left   | KEY 37\x0D       |
| Menu Navigation Menu   | KEY 03\x0D       |
| Menu Navigation Right  | KEY 38\x0D       |
| Menu Navigation Up     | KEY 35\x0D       |
| Rear Mode Off          | HREVERSE OFF\x0D |
| Rear Mode On           | HREVERSE ON\x0D  |
| Touch Unit Control Off | ITRLC 00\x0D     |



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|                       |              |
|-----------------------|--------------|
| Touch Unit Control On | ITRLC 01\x0D |
| Volume 0              | VOL 0\x0D    |
| Volume 20             | VOL 243\x0D  |

## Appendix B. Query Commands

|                    |                |
|--------------------|----------------|
| A/V Mute           | MUTE?\x0D      |
| Aspect Ratio       | ASPECT?\x0D    |
| Ceiling Mode       | VREVERSE?\x0D  |
| Closed Caption     | CCAP?\x0D      |
| Freeze             | FREEZE?\x0D    |
| Input              | SOURCE?\x0D    |
| Lamp Mode          | LUMINANCE?\x0D |
| Lamp Usage         | LAMP?\x0D      |
| Power              | PWR?\x0D       |
| Rear Mode          | HREVERSE?\x0D  |
| Touch Unit Control | ITRLC?\x0D     |
| Volume             | VOL?\x0D       |